Exact indirect-gap boundary in a flux-one-third Harper–Hofstadter model with a period-three mass

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Abstract

For a square-lattice Harper–Hofstadter Hamiltonian with flux $2\pi/3$, anisotropic hoppings $t_x = 1$ and $t_y = r$, and a magnetic-cell potential $m(1, 0, -1)$, a direct band touching does not in general coincide with the fixed-filling metal–insulator boundary. We determine both indirect gaps over $1/2 \le r \le 2$ and all real m by an exact computer-assisted calculation. The gaps at one-third and two-thirds filling are equal. Their common boundary has an algebraic cusp at $r_c = [(69 - 11\sqrt{33})/8]^{1/2} = 0.852189\ldots$. Below the cusp the only gap zeros are the direct lines $m = \pm\sqrt{3}r$. Above it, each direct line lies strictly inside an indirect band-overlap window whose two edges are given by an exact one-parameter algebraic representation. A chemical-potential criterion and Fejér–Riesz factorization reduce the global optimization over the Brillouin torus to rational polynomial identities. Sturm sequences exhaust the physical and exceptional sectors, establishing that no further extrema are missed. The isolated-band Chern tuple is $(+1, -2, +1)$ for $|m| < \sqrt{3}r$ and $(0, 0, 0)$ for $|m| > \sqrt{3}r$, even where an indirect overlap makes the fixed-filling system metallic.

I. INTRODUCTION

At rational magnetic flux, the Harper equation becomes periodic and the tight-binding spectrum splits into a finite set of magnetic subbands [1]. Their quantized Hall response is encoded by Chern numbers [2]; the flux-one-third spectrum, its band topology, and cold-atom realizations are reviewed in Ref. [3]. Experimental superlattices can tune both the topology and separation of Hofstadter bands [4]. These developments make it important to distinguish two notions that coincide only in special models: isolation of adjacent bands at every common momentum and a positive *indirect* gap between their global extrema.

This distinction matters at fixed filling. An individual band can remain isolated and carry a well-defined Chern number even when its maximum exceeds the minimum of the next band at another momentum. The system is then an indirect metal rather than a band insulator. Phase diagrams based solely on a minimum same-momentum separation do not detect this overlap. Related commensurate Aubry–André–Harper models, for example, have been classified using direct-gap observables while varying diagonal and off-diagonal phason

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modulations [5]; that Hamiltonian and observable differ from the static mass deformation considered here.

We solve the global problem for the three-band flux-one-third Hamiltonian with the period-three onsite pattern $(m, 0, -m)$. The direct closures $m = \pm\sqrt{3}r$ provide a useful baseline, but they are not the whole answer. For sufficiently large anisotropy they sit inside finite indirect-overlap windows. The exact boundary of those windows has a cusp and two algebraic branches. The central technical point is to replace two independent global band optimizations by the existence of a common chemical potential. At a boundary that condition becomes positivity of a degree-three trigonometric polynomial, allowing a complete spectral-factor reduction and an exact Sturm audit of every sector.

Section II fixes the model and derives the momentum and spectral symmetries. Section III states the complete boundary and its extremizers. Sections IV–VI give the factorization and completeness proof, and Sec. VII gives the Chern classification and physical limits.

II. MODEL, GLOBAL GAPS, AND EXACT REDUCTIONS

We consider one noninteracting particle on an infinite square lattice with lattice spacing one. In Landau gauge the real-space Hamiltonian is

$$\begin{aligned}\widehat{H} = & -\sum_{x,y}\left(c^{\dagger}_{x+1,y}c_{x,y} + re^{2\pi \mathrm{i}x/3}c^{\dagger}_{x,y+1}c_{x,y} + \mathrm{H.c.}\right) \\ & + \sum_{x,y} V_{x \bmod 3} c^{\dagger}_{x,y}c_{x,y}, \qquad (V_{-1}, V_0, V_{+1}) = (m, 0, -m). \end{aligned} \tag{1}$$

The energy unit is $t_x = 1$, $r = t_y/t_x \in [1/2, 2]$, and m is real. With $\theta = k_y$ and magnetic-cell momentum $\kappa = 3k_x$, both taken in $[-\pi, \pi)$, the ordered sublattice basis $s = (-1, 0, +1)$ gives

$$H_{r,m}(\theta, \kappa) = \begin{pmatrix} m - 2r\cos(\theta - 2\pi/3) & -1 & -e^{-\mathrm{i}\kappa} \\ -1 & -2r\cos\theta & -1 \\ -e^{\mathrm{i}\kappa} & -1 & -m - 2r\cos(\theta + 2\pi/3) \end{pmatrix}. \tag{2}$$

This deformation is our model definition; it is not attributed to the undeformed source Hamiltonian of Refs. [1, 3].

Let $E_1 \le E_2 \le E_3$ be the ordered eigenvalues. At fillings $n/3$, $n = 1, 2$, define

$$G_n(r, m) = \min_{\mathbb{T}^2} E_{n+1} - \max_{\mathbb{T}^2} E_n, \qquad d_n(r, m) = \min_{\mathbb{T}^2}(E_{n+1} - E_n). \tag{3}$$

Thus $G_n > 0$ is the band-insulator condition, whereas $d_n > 0$ only says that the two bands are isolated at each momentum.

The matrix is traceless, and direct expansion gives the Chambers-type cubic

$$\begin{aligned} P(E) &= \det(E - H) = E^3 - AE - D, \\ A &= m^2 + 3r^2 + 3 - 2\sqrt{3}mr\sin\theta, \\ D &= 2m^2 r\cos\theta - 2\sqrt{3}mr^2\sin 2\theta - 2r^3\cos 3\theta - 2\cos\kappa. \end{aligned} \tag{4}$$

For a simple ordered root at fixed θ,

$$\frac{\partial E_j}{\partial D} = \frac{1}{3E_j^2 - A}. \tag{5}$$

The denominator is positive for the two outer roots and negative for the middle root. Continuity extends the ordering through the already classified double roots. Because D depends on κ only through $-2\cos\kappa$, all relevant κ extrema lie at an endpoint:

$$\begin{aligned} G_1 &= \min_\theta E_2(\theta,\pi) - \max_\theta E_1(\theta,\pi), \\ G_2 &= \min_\theta E_3(\theta,0) - \max_\theta E_2(\theta,0). \end{aligned} \tag{6}$$

No finite momentum mesh enters this reduction.

The momentum-independent unitary

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{7}$$

obeys

$$H_{r,m}(\pi - \theta, \pi - \kappa) = -U H_{r,m}(\theta,\kappa) U^\dagger. \tag{8}$$

It follows that $E_j(\pi - \theta, \pi - \kappa) = -E_{4-j}(\theta,\kappa)$ and hence

$$\boxed{G_1(r,m) = G_2(r,m) \equiv G(r,m).} \tag{9}$$

The characteristic polynomial is also invariant under $(m,\theta) \mapsto (-m,-\theta)$, so G is even in m.

For reference, the only direct degeneracies are

$$(m,\theta) = (+\sqrt{3}r, +\pi/2), \quad (-\sqrt{3}r, -\pi/2), \tag{10}$$

with $\kappa = 0$ for the upper gap and $\kappa = \pi$ for the lower gap. Indeed, subtracting a double eigenvalue from a 3×3 Hermitian matrix makes it rank one. The three off-diagonal magnitudes in Eq. (2) are all unity, so rank-one factorization forces the three shifted diagonal entries to agree; tracelessness then forces the original diagonals to vanish. Solving those equations gives Eq. (10). The remaining zero-diagonal ring has polynomial $E^3 - 3E + 2\cos\kappa$, whose discriminant vanishes only at $\kappa = 0, \pi$.

III. COMPLETE INDIRECT-GAP BOUNDARY

We first give a square-root-free parametrization of the positive-mass indirect boundary. For $X \in (3/4, 1)$ define

$$A_X = 8X^2 - 4X - 1, \qquad B_X = 40X^3 - 36X^2 + 6X - 1,$$
$$V(X) = \frac{3(2X-1)^2 A_X}{4B_X}, \tag{11}$$

and

$$e_0(X) = -\frac{(2X-1)(16X^3 - 60X^2 + 36X - 1)}{2B_X}, \tag{12}$$
$$k_0(X) = -\frac{(X-1)(2X-1)(2X+1)^2(4X-3)(4X-1)^2}{6A_X B_X^2} \times (28X^2 - 20X + 1)^2, \tag{13}$$
$$a(X) = \frac{k_0(X)V(X)}{X}. \tag{14}$$

Let $Y_+(X)$ be the unique positive root satisfying $e_0 Y \geq 1$ of

$$a(X)Y^3 - 3e_0(X)Y + 2 = 0. \tag{15}$$

Away from the cusp the inequality is strict. Set

$$r(X) = Y_+(X)\sqrt{\frac{V(X)}{X}}, \tag{16}$$
$$u(X) = \frac{4\sqrt{V(X)(1-X)}(4X-1)}{3(2X-1)},$$
$$m(X) = \sqrt{3}\, u(X) r(X), \qquad E(X) = e_0(X) Y_+(X). \tag{17}$$

The two branches meet at

$$X_c = \frac{7 + \sqrt{33}}{16}, \qquad r_c = \sqrt{\frac{69 - 11\sqrt{33}}{8}} = 0.852189157998510\ldots, \tag{18}$$

where $u = 1$, $E = 1$, and $m = \sqrt{3}r_c$. The function $r(X)$ decreases strictly from infinity to r_c on $(3/4, X_c)$ and increases strictly from r_c to infinity on $(X_c, 1)$. Therefore, for every $r > r_c$, there are exactly two inverse values

$$X_<(r) \in (3/4, X_c), \qquad X_>(r) \in (X_c, 1). \tag{19}$$

Define

$$m_+(r) = m[X_<(r)], \qquad m_-(r) = m[X_>(r)]. \tag{20}$$

They obey $0 < m_- < \sqrt{3}r < m_+$.

The main result is the following complete sign classification:

$$\begin{aligned} &\tfrac{1}{2} \le r \le r_c: \; G > 0 \text{ for } m \neq \pm\sqrt{3}r, \quad G = 0 \text{ at } m = \pm\sqrt{3}r; \\ &r_c < r \le 2: \; G < 0 \iff m_-(r) < |m| < m_+(r), \\ &\qquad\qquad G = 0 \iff |m| = m_-(r) \text{ or } m_+(r), \end{aligned} \tag{21}$$

with $G > 0$ otherwise. Thus the relative boundary of the insulating set consists of the two direct segments $m = \pm\sqrt{3}r$ for $1/2 \le r \le r_c$, joined at the cusps to the four mass-reflected indirect branches $m = \pm m_\pm(r)$ for $r_c \le r \le 2$. For $r > r_c$ the direct line is a Chern-changing band touching *inside* the metallic window and is not a boundary of the fixed-filling insulating set.

Figure 1 shows this distinction. Selected evaluations of the exact parametrization are listed in Table I; the displayed digits are not used in the proof.

The global extremizers are also fixed by the parametrization. For positive m, set

$$\phi = -\arccos\sqrt{X}, \qquad \delta = \arccos[-\sqrt{V(X)}], \qquad \theta_c = \phi + \delta, \quad \theta_v = \phi - \delta. \tag{22}$$

TABLE I. Positive-mass indirect boundaries. Values are numerical evaluations of the exact parametrization.

r	$m_-(r)$	$m_+(r)$
0.90	1.491168750337	1.732042744620
1.00	1.522145273973	2.269195105327
1.25	1.591352307724	3.569738391484
1.50	1.646066824501	4.789938828072
2.00	1.721949718751	7.074701110742

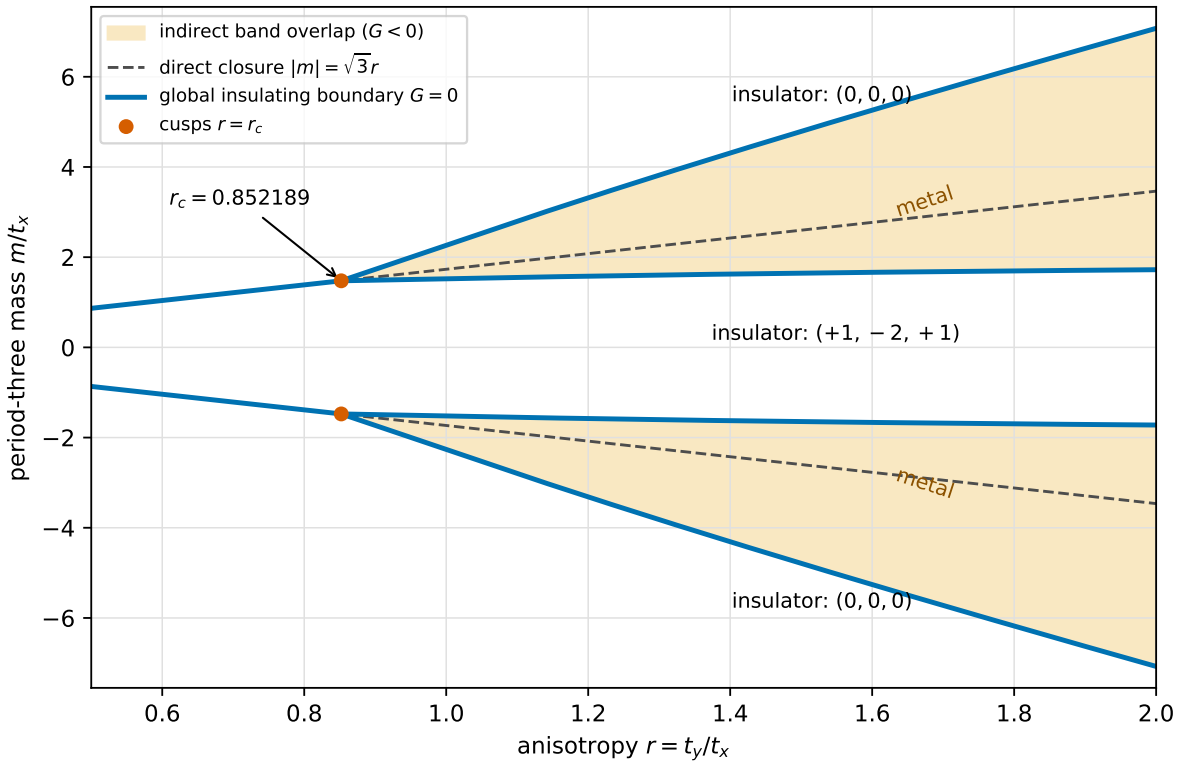


FIG. 1. Global fixed-filling diagram for either 1/3 or 2/3 filling. Blue curves are the exact $G = 0$ boundary, orange regions have indirect overlap $G < 0$, and unshaded regions have $G > 0$. The dashed direct closures continue through the metallic windows and change the isolated-band Chern tuple there. The vector plot is a deterministic evaluation of Eqs. (11)–(20); AI assisted in drafting the plotting code, whose coordinates were checked against the exact formulas and Table I.

At filling 2/3, $(\theta_c, 0)$ is the E_3 minimum and $(\theta_v, 0)$ is the E_2 maximum. At filling 1/3, the corresponding conduction and valence momenta are $(\pi - \theta_v, \pi)$ and $(\pi - \theta_c, \pi)$. Negative mass negates every θ coordinate. On a direct segment the contact is at $\theta = \mathrm{sgn}(m)\pi/2$, with $\kappa = 0$ for 2/3 filling and $\kappa = \pi$ for 1/3 filling. At the cusp Eq. (22) includes the direct contact and a distinct-momentum global extremum, so the two kinds of closure meet without being conflated.

IV. CHEMICAL-POTENTIAL CRITERION

It remains to prove that Eqs. (21) and (22) are exhaustive. Put $q = m/\sqrt{3}$ and initially take $m \geq 0$. At $\kappa = 0$, Eq. (4) becomes

$$f_{E,r,q}(\theta) = E^3 - 3E(1 + r^2 + q^2) + 2 + 6Eqr\sin\theta - 6q^2 r\cos\theta + 6qr^2\sin 2\theta + 2r^3\cos 3\theta. \tag{23}$$

For general $K = \cos\kappa$, $P_K(E,\theta) = f_{E,r,q}(\theta) + 2(K-1) \leq f_{E,r,q}(\theta)$.

Every Bloch matrix has $E_3 \geq 1$ and $E_1 \leq -1$. To see this, note that $\operatorname{tr} H^2 = 2A \geq 6$. For a traceless ordered triple whose largest entry is M, direct optimization of the other two entries gives $\sum_j E_j^2 \leq 6M^2$; applying the same statement to $-H$ gives both bounds. For $E \geq 1$, the condition $P_K(E,\theta) \leq 0$ is therefore equivalent to $E_2(\theta,\kappa) \leq E \leq E_3(\theta,\kappa)$. Combining this fact with Eq. (6) yields the exact intersection criterion

$$G_2 \geq 0 \quad \Longleftrightarrow \quad \exists E \geq 1 \text{ such that } f_{E,r,q}(\theta) \leq 0 \text{ for all } \theta. \tag{24}$$

Strict inequalities characterize $G_2 > 0$. At a genuinely indirect zero, $h(\theta) = -f(\theta)$ is a nonnegative degree-three trigonometric polynomial with two distinct zeros, the conduction and valence momenta. This turns the global band comparison into an exact positivity problem.

V. SPECTRAL FACTORIZATION AND ALGEBRAIC REDUCTION

Let $z = e^{\mathrm{i}\theta}$ and let z_c, z_v be the two unit-circle zeros of h. Fejér–Riesz factorization gives

$$h(\theta) = \left|\alpha(z - z_c)(z - z_v)(z - \zeta)\right|^2. \tag{25}$$

The third Fourier coefficient is $-r^3$. Replacing the spectral factor by its reciprocal if necessary permits

$$\zeta = \frac{\rho}{z_c z_v}, \qquad \rho \geq 1, \qquad |\alpha|^2 = \frac{r^3}{\rho}, \qquad t = \rho + \rho^{-1} \geq 2. \tag{26}$$

Write

$$\phi = \frac{\theta_c + \theta_v}{2}, \quad \delta = \frac{\theta_c - \theta_v}{2}, \quad v = \cos\delta, \quad u = \frac{q}{r}, \quad e = \frac{E}{r}. \tag{27}$$

Direct autocorrelation of Eq. (25) and matching the second and first Fourier coefficients of Eq. (23) gives

$$4v\cos\phi + t\cos 2\phi = 0, \qquad 4v\sin\phi - t\sin 2\phi = 3u, \tag{28}$$

$$\operatorname{Re} W = -3u^2, \qquad \operatorname{Im} W = -3ue, \tag{29}$$

where

$$W = (2+4v^2)e^{2\mathrm{i}\phi} + 4tve^{-\mathrm{i}\phi} + e^{-4\mathrm{i}\phi}. \tag{30}$$

The constant autocorrelation is

$$H = 2(1+2v^2)t + 8v\cos 3\phi, \tag{31}$$

and its matching equation is

$$[H + e^3 - 3e(1+u^2)]r^3 - 3er + 2 = 0. \tag{32}$$

No energy equation or stationarity condition has been squared in this derivation.

The representation redundancy $(\phi, v) \mapsto (\phi+\pi, -v)$ allows $\cos\phi > 0$. In the positive-mass two-angle sector the equations then fix $\sin\phi < 0$ and $v < 0$. Put $X = \cos^2\phi$ and $V = v^2$. Solving the two linear equations (28), reducing Eq. (29) modulo $\cos^2\phi + \sin^2\phi = 1$, and eliminating even powers produces a single primitive relation,

$$\begin{aligned} 0 = {} & -160VX^3 + 144VX^2 - 24VX + 4V \\ & + 96X^4 - 144X^3 + 60X^2 - 3. \end{aligned} \tag{33}$$

It is linear in V and solves uniquely to Eq. (11). The remaining matches give Eqs. (12)–(15). Thus the displayed one-parameter curve is not a fit to a momentum scan; it is the generic real solution of the full Fourier matching ideal.

At a zero of P, the sign of $P_E = 3E^2 - A$ distinguishes the ordered band. On both open branches the two factor zeros obey

$$P_E(\theta_c) > 0 > P_E(\theta_v). \tag{34}$$

Together with $h \geq 0$, this proves that they are respectively the global E_3 minimum and E_2 maximum, rather than two extrema of one band.

VI. EXACT COMPLETENESS CERTIFICATE

We summarize the exact sector and monotonicity certificate; Appendix A gives its decisive polynomials. The first matching equations imply

$$t^2 = \frac{16XV}{(2X-1)^2}, \qquad t^2 - 4 = \frac{4(1-X)(4X-1)^2}{B_X}. \tag{35}$$

The requirements $t \geq 2$ and $0 < V < 1$ first reduce the generic sector to the part above the sole relevant zero of B_X. In the remaining possible interval below $X = 3/4$, $e_0 > 0$ while the cubic coefficient corresponding to k_0 is nonpositive. The constant-match cubic is strictly decreasing for positive r, and at $E = 1$ it is already negative. Its unique positive root therefore has $E < 1$ and is excluded by Eq. (24). Every generic physical contact consequently satisfies

$$\frac{3}{4} < X < 1. \tag{36}$$

On this interval Eq. (15) has one negative and two positive roots, with exactly one root satisfying $E = e_0 Y \geq 1$ except for their coalescence at the cusp. The exact sign identity

$$\begin{aligned} e_0^3 - a = {} & \frac{3(2X-1)^3 A_X (16X^2 - 14X + 1)^2}{8X B_X^3} \\ & \times (32X^4 - 8X^3 - 36X^2 + 22X - 1) \end{aligned} \tag{37}$$

locates that root without a numerical branch choice. Rational Sturm sequences show that all displayed factors have the required fixed signs.

For monotonicity, impose Eq. (15) and differentiate $r = Y\sqrt{V/X}$. After denominator and endpoint factors are removed, the stationarity resultant contains only

$$(16X^2 - 14X + 1)^2 S_{12}(X). \tag{38}$$

The first factor has the physical root X_c in Eq. (18). An exact Sturm count finds one root of S_{12} in $(3/4, 1)$, isolated in $(0.769, 0.770)$; substituting its exact rational expression for Y proves $Y < 0$. It belongs to the negative cubic root and cannot make the physical branch stationary. Endpoint limits give $r(X) \to \infty$ on both sides of the interval, proving the stated two-branch monotonicity.

No division sector is discarded silently. If $\cos\phi = 0$, the first matching equation would give $t = 0$; $X = 1/2$ or $v = 0$ gives $t = \sqrt{3} < 2$; $\sin\phi = 0$, $u = 0$, or $t = 2$ reduces to the

massless sector. At $m = 0$ the choice $E = \sqrt{1 + r^2}$ lies strictly inside the gap because

$$(1 + r^2)^3 - (1 + r^3)^2 = r^2(3r^2 - 2r + 3) > 0. \tag{39}$$

A coincident pair of contact angles, a single unit-circle zero, or a unit third factor reduces to either this massless endpoint or the direct degeneracies already classified in Eq. (10). Negative mass is covered by exact reflection.

Exact algebraic-root isolation at one rational point on each open branch proves Eq. (34). Its sign cannot change before the cusp because $P_E = 0$ would be an additional direct degeneracy, already excluded. Finally, the exhaustive zero classification and continuity determine the signs in Eq. (21): the undeformed and large-mass components have positive gap, while direct substitution on $m = \sqrt{3}r$ gives a negative sample for $r > r_c$. This closes the global argument, including parameter edges and exceptional sectors.

All decisive polynomial operations use exact integer or rational arithmetic, with the explicit quadratic field $\mathbb{Q}(\sqrt{33})$ at the cusp. The machine-readable certificate reconstructs the Laurent coefficients directly from Eq. (2), derives Eq. (33) without importing $V(X)$, factors the resultants, performs the Sturm counts, and asserts every exceptional-sector identity. Floating-point diagonalization is used only as an independent replay and for Fig. 1; it is not the source of exhaustiveness.

OpenAI Codex (a GPT-5-based agent, September 2026 service version) assisted with problem formulation, literature synthesis, symbolic derivations, and generation and debugging of the verification code. It was directed by the fixed Hamiltonian and explicit exact-arithmetic assertions. Outputs were cross-checked using two independently constructed Bloch representations, coefficient-by-coefficient Laurent reconstruction, rational Sturm sequences, and a separately implemented numerical replay. These computational checks are not a substitute for human peer review; final human scientific review is required before submission.

VII. CHERN LABELS AND PHYSICAL SCOPE

Equation (10) proves that the three individual bands can change Chern number only on $m = \pm\sqrt{3}r$. The central parameter region is connected to $r = 1, m = 0$, whose positive-flux, energy-ordered tuple is $(+1, -2, +1)$ [3]. It is also reproduced by converged link-variable

and independent Kubo-curvature calculations. Each outer region can be deformed to large $|m|$ without crossing a direct closure. For $|m| > 4(r+1)$, scale all hoppings by $\lambda \in [0,1]$. Gershgorin intervals of radius $2\lambda(r+1)$ around the onsite energies $-m, 0, +m$ remain disjoint. At $\lambda = 0$ the projectors are constant atomic-sublattice projectors, so every Chern number vanishes. Hence

$$(C_1, C_2, C_3) = \begin{cases} (+1, -2, +1), & |m| < \sqrt{3}r, \\ (0, 0, 0), & |m| > \sqrt{3}r. \end{cases} \tag{40}$$

Complex conjugation or orientation reversal changes the nonzero tuple to $(-1, +2, -1)$ and leaves the zero tuple unchanged.

The dashed portions of Fig. 1 illustrate why Chern labels and fixed-filling insulation must be reported separately. On either side of a direct closure inside an orange window, the bands are again individually isolated and Eq. (40) applies, but $G < 0$ and neither filling is a band insulator.

The result is restricted to the infinite, clean, noninteracting single-particle Hamiltonian in Eq. (2). It does not include interactions, disorder, trapping, edges, finite-size corrections, or an experimental error budget, and it is not a claim about a many-body thermodynamic transition. The exact certificate is computer assisted rather than formalized in a proof assistant. A bounded comparison with period-three superlattice, generalized Harper, and commensurate Aubry–André–Harper literature found no identical static-mass global boundary, but this non-detection is not an exhaustive priority claim and warrants broader human literature review.

VIII. CONCLUSION

The global insulating boundary of this flux-one-third mass-deformed Harper–Hofstadter model is qualitatively different from its direct-gap phase diagram. An exact chemical-potential factorization reveals an algebraic cusp at r_c and, above it, two mass edges surrounding each direct topological closure. Exact sector elimination and Sturm counts prove that these are all the edges over the full Brillouin torus and parameter domain. The resulting diagram cleanly separates the Chern topology of isolated bands from the existence of a fixed-filling band insulator.

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OpenAI Codex (a GPT-5-based agent, September 2026 service version) also assisted with literature organization, manuscript drafting, and deterministic data-visualization code. The scientific uses, direction, and computational cross-checks are described in Sec. VI. The human author must review and refine all content, verify the disclosure against the current APS policy and applicable service terms, and accept responsibility before any submission. No external peer review or editorial assessment has occurred.

DATA AVAILABILITY

The complete LaTeX source, exact-arithmetic verification scripts, generated boundary data, figure-generation code, software versions, and regeneration instructions are included in the accompanying reproducibility archive. The CSV data are a sampled visualization of the exact equations, not the definition of the boundary.

Appendix A: Decisive stationary polynomial

For completeness, the degree-12 factor in Eq. (38) is

$$\begin{aligned} S_{12}(X) = {} & 602112X^{12} - 1505280X^{11} + 1069824X^{10} + 184576X^{9} \\ & - 524928X^{8} + 195264X^{7} - 23328X^{6} + 8736X^{5} - 3528X^{4} \\ & + 228X^{3} - 45X^{2} + 15X - 1. \end{aligned} \tag{A1}$$

At its sole root in $(0.769, 0.770)$, the simultaneous stationary solution is

$$Y = -\frac{2B_X P_5(X)}{X(2X-1)P_7(X)}, \tag{A2}$$

where

$$P_5 = 224X^5 - 296X^4 + 76X^3 + 10X^2 + 5X - 1, \tag{A3}$$

$$\begin{aligned} P_7 = {} & 3584X^7 - 23296X^6 + 38688X^5 - 24304X^4 \\ & + 5024X^3 - 96X^2 + 182X - 25. \end{aligned} \tag{A4}$$

Exact root counts show that B_X, P_5, P_7 have no zero in the isolating interval and have the signs that make $Y < 0$. The full numerator resultant, up to a nonzero rational factor, is

$$(4X-1)^2X^3(2X-1)^3A_X(16X^2-14X+1)^2 \times (28X^2-20X+1)^2B_X^9S_{12}(X). \qquad \text{(A5)}$$

The factors outside $(3/4, 1)$ or already known to be nonzero on that interval are retained here to make the division structure explicit.

Appendix B: Undeformed benchmark

At $r = 1, m = 0$, Eq. (4) reduces to

$$P(E) = E^3 - 6E + 2[\cos(3\theta) + \cos\kappa]. \qquad \text{(B1)}$$

Because the bracket fills $[-2, 2]$, the three exact band ranges are

$$E_1 \in [-1-\sqrt{3}, -2], \quad E_2 \in [1-\sqrt{3}, \sqrt{3}-1], \quad E_3 \in [2, 1+\sqrt{3}]. \qquad \text{(B2)}$$

Both direct gaps are therefore $3-\sqrt{3}$. This reproduces a source-anchored limit independently of the indirect-boundary calculation.

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